

Few-shot Bioacoustic Event Detection

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Abstract—Few-shot bioacoustic sound event detection aims to identify animal vocalizations in long and noisy recordings using only a small number of labeled examples. This problem is central to DCASE Task 5, where each recording represents an independent detection task with limited positive annotations and a large amount of background audio. In this work, we reproduce and extend a recent Regularized Supervised Contrastive Learning (RSCL) transfer-learning framework for few-shot detection. We evaluate different fine-tuning strategies, including prototypical loss and cross-entropy classification, with and without backbone freezing, on a reduced validation split derived from the DCASE 2024 dataset. Our results show that RSCL pretraining combined with prototypical fine-tuning consistently outperforms the official baseline for DCASE 2024 Task5 and our cross-entropy-based implementation. Further improvements are obtained by increasing the number of augmented views during contrastive pretraining, with a three-view configuration achieving the best overall F1-score of 62.417%. These findings highlight the importance of transferable representations and distance-based adaptation in few-shot bioacoustic detection under realistic data constraints.

I. INTRODUCTION

A. Background

Bioacoustics studies biological sound produced by animals (e.g., bird song and mammal vocalisations) and recorded in natural environments [1], where the acoustic clip may include multiple sounds incorporating non-target species and background noise [2]. Sound event detection (SED) addresses the problem of identifying when an acoustic event occurs by estimating onset and offset times, commonly using time-frequency representations such as log-mel spectrograms [2]. Deep learning techniques have been used to handle the complexity and challenges of acoustic data in bioacoustics as to extract meaningful features to detect the sound events. However, the lack of labeled data is still considered an obstacle for applying these techniques [1]. Thus, few-shot learning (FSL) emerged as a solution [3]. Few-shot learning refers to learning or adapting from only a small number of labeled examples per target class (“shots”), typically organized into a labeled support set and an unlabeled query set [4]. DCASE Task 5 instantiates this setting for few-shot bioacoustic event detection (FSBED) where each recording defines an independent task with a single class of interest, a small labeled support set, and a query portion where additional event instances must be detected over time [5].

B. Motivation

In the few-shot learning setting, where only a few labeled examples are available for new classes, Snell et al. [4] proposed Prototypical Networks (ProtoNet). The idea is to learn an embedding, compute one prototype per class as the average of its support embeddings, and classify each query by the nearest prototype. This approach mimics the idea of how clustering works and is trained with episodes that copy the test setup (N-way, K-shot), similar to the episodic training used in Matching Networks by Vinyals et al. [6].

Building on this foundation, a lot of the recent implementations for few-shot bioacoustic sound event detection adopt prototypical learning as a baseline because the setting is similar, along with additional issues in the dataset like uneven or long recordings, noisy backgrounds, and big domain shifts across devices and habitats. Liang et al. [7] showed that this standard prototypical setup can break under these conditions so he suggested refining it by building positive prototypes from event examples and negative prototypes from background segments, using hard negative sampling to pick the most confusing background segments rather than random ones. This simple change improved the results over the vanilla prototypical approach. While ProtoNets and such related meta-learning approaches have been recently among the most commonly used frameworks for few-shot audio, evidences in computer vision show that simpler transfer-learning setups can often do better than meta-learning methods in FSL [8]. This shifts the emphasis to designing transferable feature extractors instead that can adapt efficiently with minimal labeled data.

One approach that fits this idea well is supervised contrastive learning. Derived from Contrastive learning (CL) which learns an embedding space that pulls similar pairs close and dissimilar pairs apart but doesn’t require labels to form the “positive” pairs as it uses different augmented views of the same sample and treats other batch samples as negatives. While CL has been widely used in the literature and has shown promising results in audio representation learning [9], Khosla et al. [10] extend contrastive learning beyond the self-supervised setting by redefining “positives” as all examples in the batch with the same class label (across multiple augmented views), using a temperature-scaled contrastive objective that encourages tight intra-class clustering and strong inter-class separation.

Building on Khosla et al.’s supervised contrastive objective [10], Moummad, Serizel, and Farrugia propose to pre-train an audio encoder (with a small projection head) so that segments from the same class end up close together in the embedding space and different classes are pushed apart; after pre-training, they drop the head and reuse the encoder for DCASE few-shot detection, first with a simple binary classifier on top [11]. In their later paper, they strengthen the pre-training by adding an information-theoretic regularizer to avoid “collapsed” or redundant embeddings and to encourage a richer feature space [12]. This regularized pre-trained feature extractor is then fine tuned at the inference using a prototypical loss and the prediction for presence/absence of the sound is done by computing the Euclidean distances between the queries and the prototypes instead of the cross-entropy [12]. This regularized contrastive pre-training framework is the main motivation for our work, and we use it as the starting point for our experiments.

C. Problem Overview

Few-shot bioacoustic sound event detection is addressed in DCASE 2024 Task 5, where the objective is to detect occurrences of a target animal sound in long field recordings using only a few annotated examples [5]. In the test setup, each audio recording defines an independent task with a single class of interest. For every task, only a small number of positive segments are annotated and provided as a support set, while the remaining portion of the recording consists largely of unlabeled background audio that may include other species, environmental noise, or silence.

A key challenge of this task lies in the mismatch between training and evaluation conditions. During training, the system has access to multi-class annotations pooled across many recordings, allowing the model to learn generalizable acoustic representations. In contrast, during validation and inference, each recording contains only one target class, and the system must adapt using only the first five annotated positive segments. In this work, we follow the official task formulation but evaluate on a reduced validation subset consisting of three classes (ME, PB, and PB24), with 11 audio files totaling approximately 5 hours and 20 minutes of audio. This reduced setup is more constrained than the full DCASE 2024 validation set and provides a challenging but controlled environment to analyze the behavior of different strategies.

D. Report Overview

The report is structured as follows. Section II presents the proposed methodology, covering dataset preparation, data augmentation strategies, model architecture, supervised contrastive pretraining, fine-tuning approaches, and inference methods. Section III and IV describe the experimental setup and results, starting with a reproduction of the DCASE 2024 Task 5 baseline on the reduced validation set, followed by an evaluation of regularized supervised contrastive learning (RSCL) configurations and augmented-view pretraining variants. Section V discusses the experimental findings, with particular emphasis on the comparison between prototypical

TABLE I: Training data augmentations. SM: Spectrogram Mixing, FS: Frequency Shift, RRTC: Random Resized Time Crop, PG: Power Gain, SA: SpecAugment (time & frequency masking), AWGN: Additive White Gaussian Noise.

Augmentation	Params	Values
SM	factor	Beta(5, 2)
FS	bands	[0, 10]
RRTC	ratio	[0.6, 1.0]
PG	factor	[0.75, 1.0]
SA	time mask / freq mask	$0.2T / 0.1F$
AWGN	std	[0, 0.1]

and cross-entropy fine-tuning, the effect of freezing the RSCL encoder backbone during finetuning, and the impact of increasing the number of augmented views during pretraining. Section VII summarizes the main findings of the study. Finally, Section VI outlines potential directions for future research.

II. METHODOLOGY

A. Dataset preparation

The training data is prepared by converting each annotated recording into a set of fixed-length log-Mel spectrogram patches, stored in a single HDF5 file. First each waveform is resampled to 22.05 kHz and converted to a Mel spectrogram using *FFT size 512*, *hop length 128*, and *128 Mel bands*, followed by a log amplitude transform to obtain log-Mel features. Then, using the corresponding annotation file, we extract training examples from every POS event interval by slicing the log-Mel spectrogram into fixed 200ms time crops (i.e., patches of size $128 \times \text{win_len}$ frames). For events longer than 200ms, we apply a sliding window with 50% overlap to generate multiple patches from the same event, so the model sees several overlapping slices from different parts of that event (e.g., beginning, middle, end) instead of just one. Empty patches (near zero energy) are discarded, and the remaining patches are normalized before being written to HDF5 along with their class labels.

B. Augmentations

The augmentations used to construct augmentation pipelines for contrastive pre-training, as well as for fine-tuning and multi-view inference are from the audio representation learning literature [9], [13], [14]. The key hyperparameters used in these augmentations are described in Table I.

During contrastive pre-training, we generate multiple augmented “views” of each log-Mel patch. We implement three augmentation pipelines:

- T1 - Spectrogram mixing followed by FS, RRTC (crop + resize), power companding, and AWGN.
- T2 - The same pipeline as T1 but without spectrogram mixing.
- T3 - Same as the T2 pipeline along with an additional SpecAugment masking (time and frequency masking) before AWGN.

The number of views used during pre-training is treated as an experimental factor as detailed in Section III. During the

validation phase, a lighter augmentation policy that uses RRTC with ratio sampled uniformly from [0.9, 1.0] of the total duration and PG with coefficient sampled uniformly from [0.9, 1.0] is implemented for fine-tuning and to create multiple query views at inference as described in [12].

C. Model Architecture

As reported in [12], we replicate the model to use a ResNet [15] based convolutional encoder with three blocks (feature maps 64/128/256). Each convolution is followed by batch normalization and a LeakyReLU activation, and max-pooling is applied after each block (2×2 , 2×2 , then 1×2). To support varying length inputs, an adaptive max-pooling layer outputs a fixed (8×1) tensor, to get a 2048 dimensional latent vector, followed by an MLP projector with a hidden layer of dimension 2048 and an output layer of dimension 512.

D. Pre Training the Feature Extractor

1) *Supervised contrastive objective*: The feature extractor (as shown in Fig 1) is pre-trained using SCL, which learns an embedding space where samples from the same class cluster together while samples from different classes are separated. Following [12], the model is composed of an encoder $f(\cdot)$ and a lightweight projection head $h(\cdot)$. During pre-training, we generate V augmented views of each input by applying a stochastic augmentation operator $\mathcal{A}(\cdot)$ multiple times, where V is treated as an experimental factor (Section III). Each view is mapped to a projected representation $z_i = h(f(\mathcal{A}(x)))$. After convergence, the projection head is discarded and the encoder f is transferred to the downstream few-shot detection stage. In practice, increasing V increases the number of positive pairs per anchor and can strengthen invariance, at the cost of additional computation.

Let $\mathcal{I}$ index all augmented views in a batch (i.e., all V views of each original sample). For an anchor index $i \in \mathcal{I}$, the positive set is defined as $\mathcal{P}(i) = \{p \in \mathcal{I} : y_p = y_i, p \neq i\}$ and the comparison set as $\mathcal{N}(i) = \mathcal{I} \setminus \{i\}$. The SCL loss is:

$$\mathcal{L}_{\text{SCL}} = \sum_{i \in \mathcal{I}} \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(z_i^\top z_p / \tau)}{\sum_{n \in \mathcal{N}(i)} \exp(z_i^\top z_n / \tau)}, [12] \quad (1)$$

τ is the temperature parameter.

2) *Regularization*: To avoid collapsed representations and encourage non-redundant embeddings, the SCL objective is augmented with a *Total Coding Rate* (TCR) regularizer from information theory which states that the coding rate is the proportion of bits that carry non-redundant information [16]. Thus, for a batch of projected features $Z = [z_1, \dots, z_b] \in \mathbb{R}^{d \times b}$, the TCR of Z is:

$$R(Z) = \frac{1}{2} \log \det \left(I + \frac{d}{b\varepsilon^2} Z Z^\top \right), [16] \quad (2)$$

where $\varepsilon > 0$ controls the precision. The training loss during pre-training then becomes:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{SCL}} - \lambda R(Z), [12] \quad (3)$$

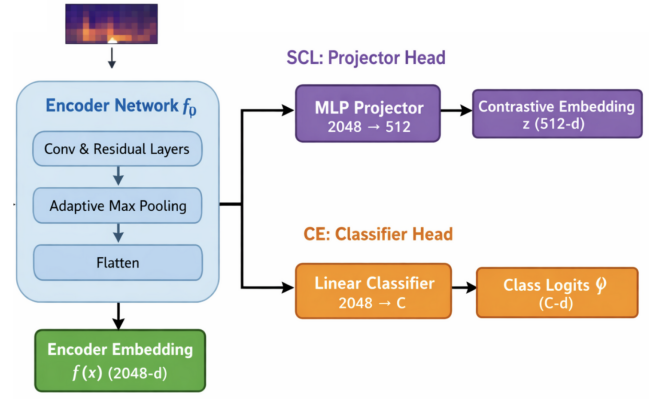


Fig. 1: Overview of the Model Architecture used for pretraining and finetuning

with $\lambda > 0$ setting the regularization strength. A large coding rate of Z encourages embeddings to use many independent dimensions (less redundancy) and is closely related to covariance-regularization objectives used in methods such as VICReg [17].

E. Fine tuning on Validation Set

For validation with finetuning, the first five annotated shots from each audio file are used as the positive support segments. While negative support segments are sampled from the intervals preceding each shot. Following Tang et al. [18], we use an adaptive window length to extract support/query segments, and we set the window hop to half the window length (50% overlap), consistent with the implementation in [12]. The two different finetuning strategies as shown in Fig 1 are:

1) *ProtoLoss*: Given the prototypes computed from the annotated support segments, we fine-tune the encoder using a softmax objective that increases similarity between an embedding and its corresponding prototype while decreasing similarity to competing prototypes [12]. This approach used for Finetune loss is similar to the ProtoNets loss [4], however, the change is that in the normalization, the positive prototype term is excluded, inspired by contrastive learning variants like DCL [19] that improve stability by removing the positive pair from the normalization. Following [12], the fine-tuning loss for an embedding z_i with target class c is:

$$\mathcal{L}_{\text{finetune}} = -\log \frac{\exp(z_i \cdot z_c)}{\sum_{c' \neq c} \exp(z_i \cdot z_{c'})} [12] \quad (4)$$

2) *Cross-entropy binary classifier loss*: Another approach, following [11], pairs the transferred pre-trained encoder with a lightweight 2-class linear classifier and fine-tunes this model to discriminate event presence (1) versus absence (0) using cross-entropy. This approach is simple and we include it in our experiments (Section III) as a straightforward baseline and to illustrate its sensitivity to the severe presence/absence

imbalance typical of bioacoustic recordings. The binary cross entropy loss minimized during validation finetuning is:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | x_i) \quad (5)$$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{y_i}(x_i))}{\exp(s_0(x_i)) + \exp(s_1(x_i))} \quad (6)$$

F. Inference

Using the same adaptive window length and hop described in Section II-E, we create query segments to predict the presence or absence of the event of interest in each file using two different methods:

1) *Nearest Prototype Classifier*: We generate multiple augmented views of each support and query segment. The embeddings obtained are averaged across views to obtain a single representation per segment. We then also compute a single positive prototype and negative prototype by averaging all the corresponding support embeddings. Finally, as described in [12] each query segment (q) is classified by comparing its embedding to the prototypes using Euclidean distance, assigning the label of the closest prototype as given in the equation below:

$$\bar{z}_i = \frac{1}{|Z_i|} \sum_{z \in Z_i} z, \quad \hat{q}_i = \arg \min_i \|q - \bar{z}_i\|_2 \cdot [12] \quad (7)$$

where Z_i denotes the set of support embeddings for class $i \in \{\text{neg}, \text{pos}\}$.

2) *Linear Classifier*: As implemented in [11], Each query segment is forwarded through the encoder-classifier network and assigned a label via an argmax over the two logits (presence vs. absence of the event). Unlike the nearest-prototype classifier, which compares query embeddings directly to class prototypes in the embedding space, this approach predicts labels using the learned linear classifier head and is used for the experimentation purposes in Section III.

Lastly, both these approaches produce a binary label sequence over time, from which event onsets and offsets are obtained by detecting transitions from negative to positive and from positive back to negative, respectively.

III. EXPERIMENTS

A. Reproducing the DCASE 2024 Task 5 Baseline on a Reduced Validation Set

As an initial experiment, we reproduced the official DCASE 2024 Task 5 baseline system and training protocol, following the baseline configuration described in [7]. Concretely, we trained the provided prototypical-network-based model on the same development training data used in the baseline paper, keeping the model parameters and feature extraction consistent with the reference setup (22.5 kHz resampling; STFT 1024 window/ 256 hop; log-mel features) [7]. While the baseline paper uses the extended validation set (“Validation set 2024”) consisting of multiple classes and long duration recordings (reported total duration 49 h 57 min

[7], our validation set is reduced as mentioned in the Problem Overview I-C.

B. Reproducing the RSCL Transfer-Learning System and Fine-tuning Variants

Based on the technical report by Moummad *et al.* [12], we reproduced the authors’ Regularized Supervised Contrastive Learning (RSCL) pretraining setup. We then evaluated two fine-tuning configurations, each with two variants (frozen and unfrozen layers) using the same hyperparameters and experimental settings described in section II.

a) *RSCL pretraining + prototypical fine-tuning*: We first ran the authors’ configuration using supervised contrastive pre-training extended with the Total Coding Rate (TCR) regularization term 1, followed by recording-wise feature adaptation with a prototypical fine-tuning loss (ProtoLoss) 4 for all the learned backbone blocks as described in sub-sections II-D and II-E1 [12].

b) *RSCL pretraining + CE fine-tuning*: We then evaluated the configuration corresponding to the cross-entropy (CE) fine-tuning variant using a linear classifier head 6 (Submission 4 in the DCASE 2023 report [11]), with finetuning all layers.

c) *RSCL pretraining + prototypical fine-tuning with all layers frozen*: To test the effect of freezing the encoder and restricting adaptation, we reran the prototypical fine-tuning configuration while freezing encoder blocks 1–3 (freezing all learned backbone blocks).

d) *RSCL pretraining + CE fine-tuning with all layers frozen*: Finally, we reproduced the CE fine-tuning configuration with a linear classifier while freezing all backbone layers (Submission 1 in the DCASE 2023 report [11]).

C. Augmented-view Pretraining Variants

In addition to the configurations above, we explored modifications to the number of augmented views used during RSCL pretraining while using the prototypical fine-tuning.

a) *Four-view RSCL pretraining*: Unlike the reference RSCL setup [12], which constructs two augmented views per training example (T1 and T2, as discussed in II-B), we generated four views using the same augmentation pipelines, applying transform T1 to views 1–2 and T2 to views 3–4.

b) *Three-view RSCL pretraining with an additional augmentation transform*: We further trained a three-view variant by adding one extra view generated by an additional transform (T3).

For both the experiments, we used batch size 64 over 150 epochs, while keeping all other hyperparameters identical to those used in III-B.

IV. RESULTS

Table II reports overall and per-class validation performance for all configurations on our ME/PB/PB24 split.

Configuration	Overall			PB			ME			PB24		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Baseline reproduction (reduced VALSet)	0.441	0.383	41.048	0.660	0.347	0.455	0.268	0.903	0.414	0.666	0.283	0.398
RSCL+ProtoLoss (no freezing)	0.733	0.528	61.358	0.649	0.362	0.464	0.783	0.904	0.839	0.784	0.551	0.647
RSCL+CE (linear, no freezing)	0.301	0.427	35.344	0.677	0.263	0.378	0.472	0.808	0.596	0.157	0.506	0.240
RSCL+ProtoLoss (all frozen)	0.597	0.535	56.411	0.552	0.391	0.458	0.515	0.981	0.675	0.784	0.492	0.604
RSCL+CE (linear, all frozen)	0.661	0.389	49.000	0.571	0.217	0.314	0.686	0.923	0.787	0.751	0.498	0.599
Four-view RSCL + ProtoLoss	0.730	0.532	61.522	0.674	0.370	0.478	0.727	0.923	0.814	0.799	0.538	0.643
Three-view RSCL (+extra transform) + ProtoLoss	0.696	0.565	62.417	0.660	0.421	0.514	0.671	0.942	0.784	0.768	0.535	0.630

TABLE II: Validation results on the ME/PB/PB24 split. Reported values are precision (P), recall (R), and F1 (F1).

V. DISCUSSION

This section discusses the observed behavior of the evaluated configurations on our validation split and summarizes how performance evolved from the initial reproductions to the best-performing variant.

A. From baseline reproduction to the transfer-learning pipeline

Reproducing the DCASE 2024 Task 5 baseline on our reduced validation set yielded an overall F1-score of 41.048% as shown in table II. Moving to the RSCL transfer-learning pipeline with prototypical fine-tuning increased the overall F1 to 61.358%, indicating that the pretrained embedding and recording-wise adaptation strategy is a strong fit for our task.

B. ProtoLoss vs. cross-entropy fine-tuning

Across our trials, prototypical fine-tuning (ProtoLoss) consistently outperformed cross-entropy (CE) fine-tuning with a linear classifier head, both with and without freezing. With no frozen layers, ProtoLoss achieved F1 score of 61.358% vs. 35.344% for CE, and with all backbone layers frozen, ProtoLoss achieved 56.411% vs. 49.000% for CE as shown in table II.

A reasonable explanation is that ProtoLoss directly optimizes a distance-based decision rule that fits the few-shot evaluation setting more than CE does. Since with ProtoLoss, query segments are compared to support-derived prototypes where no need to estimate a separate parametric decision boundary from very limited labelled examples. However, CE with a linear head suffers from class imbalance and negative-sampling choices while trying to fit a separating hyperplane from a small and potentially noisy set of positive segments together with many background segments. This difference can be shown with some subsets, where for example, CE with freezing produces consistently low F1 on PB (0.314) and ME (0.787), while ProtoLoss maintains high recall on ME (0.904) and substantially higher overall balance between precision and recall (0.839 F1).

C. Why CE improves when the backbone is frozen

With CE fine-tuning, freezing the encoder backbone during fine-tuning improved the overall F1 from 35.344% to 49.000%. This behavior is expected because trying to update

a backbone with a linear head classifier using limited labelled examples can lead to over-adaptation. The representations can shift during training trying to separate the sampled negatives from the few positives leading to poor generalization especially when the background is diverse. On the other hand, freezing the backbone constrains learning to the linear head, effectively reducing the number of trainable parameters and limiting representation shift, which can yield better generalization under small support set. The precision–recall pattern shown in table II supports this where the frozen CE configuration shows relatively high precision (0.661) but low recall (0.389), indicating a conservative classifier that avoids false positives at the expense of missed detections.

D. Effect of freezing for ProtoLoss

Freezing the backbone layers during ProtoLoss finetuning reduced the overall F1 from 61.358% to 56.411%. This is reasonable since unlike CE, ProtoLoss uses the encoder to shape the embedding geometry around the support examples. So, allowing adaptation (no freezing) with support segments capturing recording-specific characteristics can improve alignment between the pretrained embedding space and the specific recording characteristics. On the contrary, when the backbone is frozen, the method relies entirely on the pretrained representation without any task-specific adjustment, which can degrade the precision–recall balance for some subsets (e.g., ME precision decreases while recall remains high)

E. Increasing the number of augmented views during RSCL pretraining

After the four setups we experimented, we found that using Regularized SCL pre-training and prototypical fine-tuning with full adaptation gave the best result, hence we built on that. So, as discussed in section III-C, we tested a four-view variant system, and this achieved a F1 score of 61.522% instead of 61.358%. Then, we tested a three-view variant with an additional augmentation transform which achieved the best score of 62.417%. The improvement the four-views variant provided over the two-views setup was due to the increase of the number of positive pairs and the decrease of the variance of the contrastive estimate rather than introducing more diversity. In contrast, the three-view configuration adds an extra view generated by a new augmentation transform, increasing both

view number and diversity, and thus leading to a higher score. This likely provides complementary invariances during pretraining and is reflected in higher overall recall compared to the two-views setup (0.565 vs. 0.528) and improved performance on the most challenging subset PB (F1 0.514 vs. 0.464) as shown in table II.

VI. FUTURE WORK

Several directions could be explored to further extend this work. One potential approach is to explore an alternative adaptive segment length strategy during inference, beyond the one explored here, following the method described in [20] to evaluate whether changing how duration variability is modeled affects the overall performance. Another direction is to further expand the multi-view supervised contrastive pretraining framework by incorporating additional and more diverse augmentation transforms.

VII. CONCLUSION

In this work, we investigated few-shot bioacoustic sound event detection under the DCASE 2024 Task 5 framework, with an emphasis on transfer learning via Regularized Supervised Contrastive pretraining. Starting from a reproduction of the official baseline on a reduced validation set, we showed that baseline prototypical methods degrade noticeably under limited validation data. In contrast, the RSCL-based pipeline provided a substantial performance improvement, demonstrating the importance of learning a robust and transferable embedding space for this task.

Across all evaluated configurations, prototypical fine-tuning consistently outperformed cross-entropy-based classification, both with frozen and unfrozen backbones. This confirms that distance-based adaptation is better suited to the few-shot detection setting, where only a small number of positive examples are available and background audio dominates.

Finally, we observed that increasing the number of augmented views during supervised contrastive pretraining improves detection performance. While a four-view configuration provided only marginal gains over the two-view setup, a three-view configuration incorporating an additional augmentation transform achieved the best overall performance, reaching an F1-score of 62.417%. Overall, these results confirm that regularized supervised contrastive pretraining combined with prototypical fine-tuning is a strong and reliable approach for few-shot bioacoustic sound event detection under realistic low-data conditions.

AUTHOR CONTRIBUTIONS

Laiba designed the overall report structure, wrote the Motivation (Section 1B), and authored the Methodology section (Section II). Hadi authored the Background (Section 1A), Experiments (Section III), and wrote the Results and Discussion (Section IV and V). Aqsa contributed the Abstract, the problem statement and report overview (Sections 1C–1D), and authored the Conclusion and Future Work (Section VI and VII).

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